

Study case: Male vs Female

Is there a wage gap?
What are the determining factors of
salary?

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Project Overview

Gender Disparities in Hiring and Political Influence

Objective

Analyze gender disparities in employment, representation and wages across industries, occupations and U.S. states

Key Questions

- How much of the gender wage gap can be explained by occupation and industry?
- Does gender remain important after accounting for other factors?
- Where are women underrepresented in the workforce?
- Does political geography have influence on the wage disparity?

1.57 M

Employed Workers

72k +

Job Postings

3.2 M

Person Records

Data Overview

Primary Dataset

Integrated Public Use Microdata Series
(IPUMS USA)

Coverage: 2024 American Community Survey

Use: Workforce representation, wage gap analysis,
clustering and machine learning models

Why it matters: Provides detailed worker-level data
needed to measure gender disparities in
employment and earnings

Secondary Dataset

Lightcast Job Postings Database

Use: To validate workforce trends against current
labor market demand

Why it matters: Connects workforce outcomes to
real-world hiring demand

Pipeline Architecture

1. Extract

IPUMS: 2024 ACS, 3.2M records
Lightcast: 2024 job postings
72.000

2. Transform & Clean

IPUMS: 1.57M employed, positive wages
Lightcast: 72.000 postings cleaned (salary, state, NAICS)

3. Load & Features

/data/raw
to
/data/processed

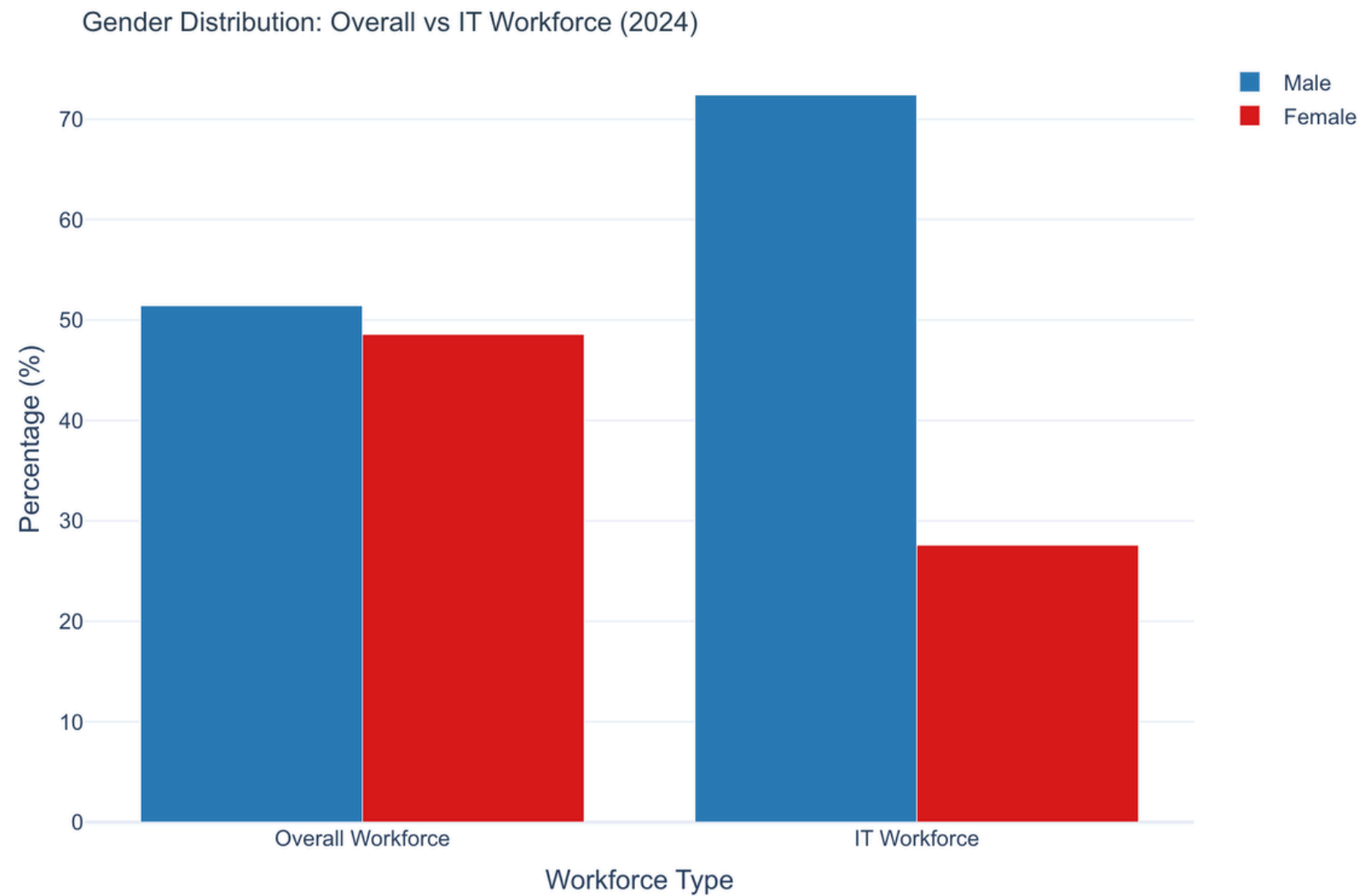
```
✓ data
  > external
  ✓ processed
    > ipums_clustered.parquet
    ◆ .gitkeep
    ■ employed_only.csv
    ≡ employed_only.parquet
    ■ employment.csv
    ■ gender_clf_confusion.csv
    ■ gender_clf_metrics.csv
    ■ gender_clf_roc.csv
```

4. Model & Serve

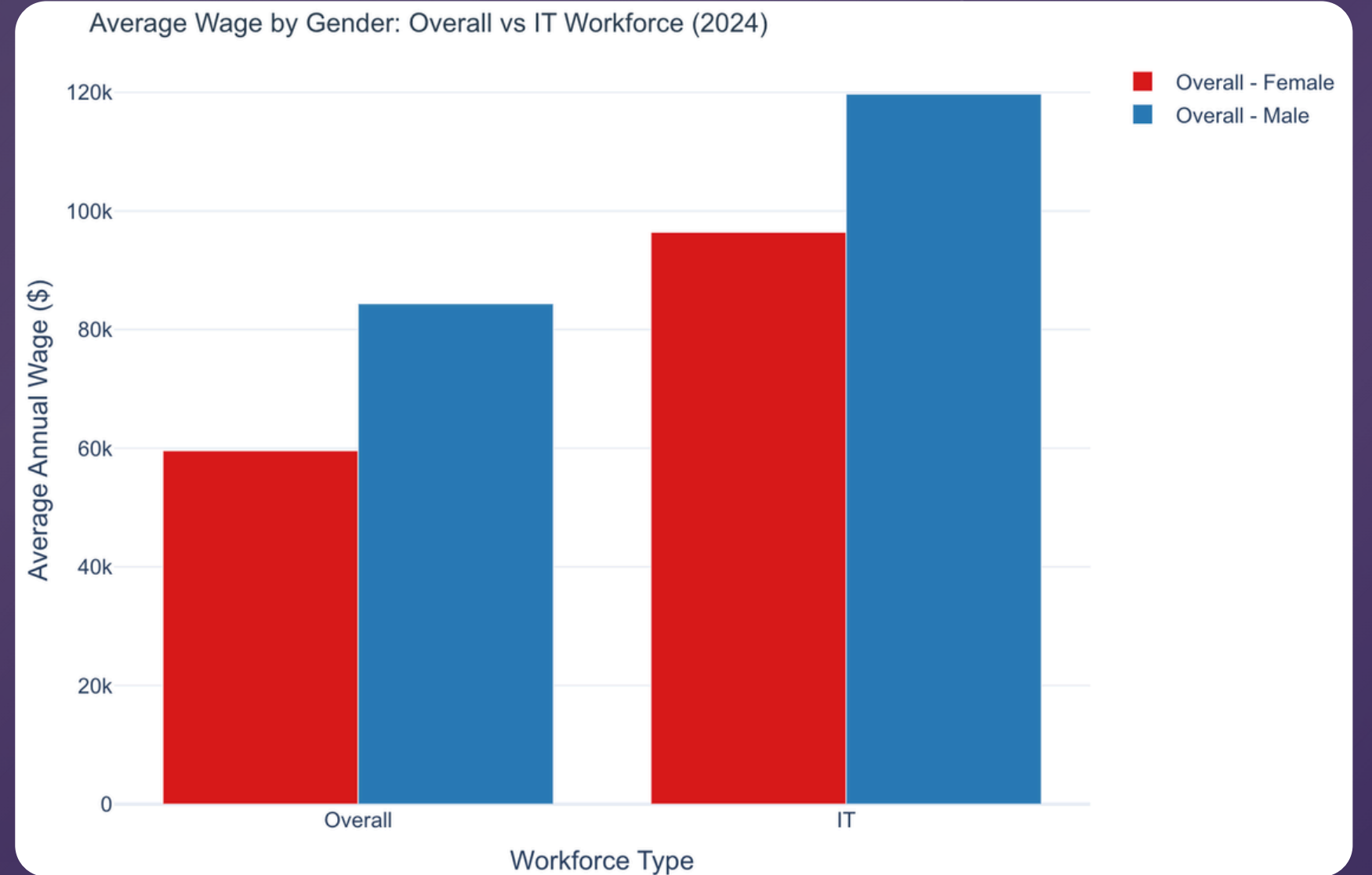
Cluster + Classification
on the IPUMS side,
→ joining the Lightcast job posting to
see wage by gender across states.

Visuals

Gender Distribution by Workforce

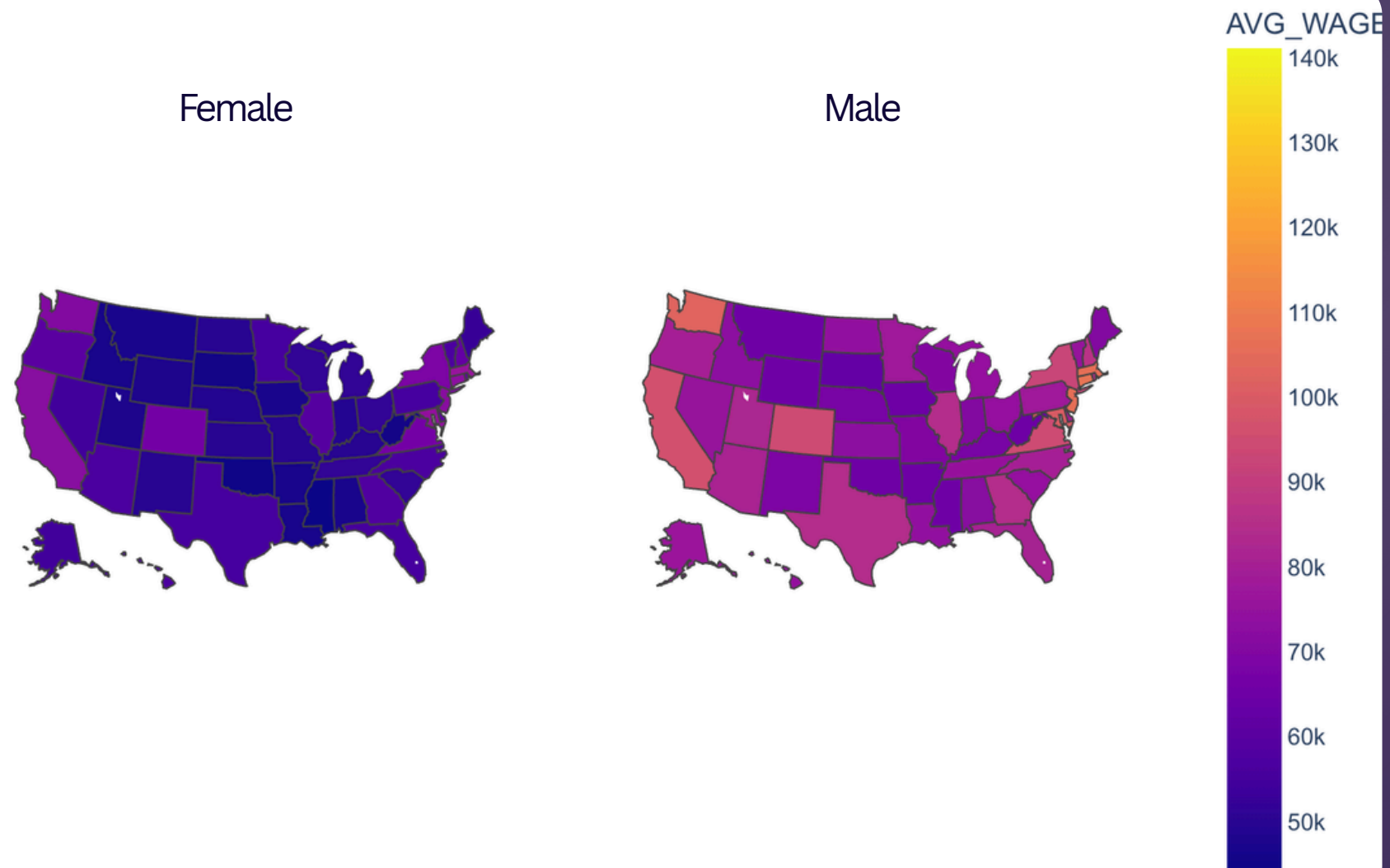


Average Wage by Gender

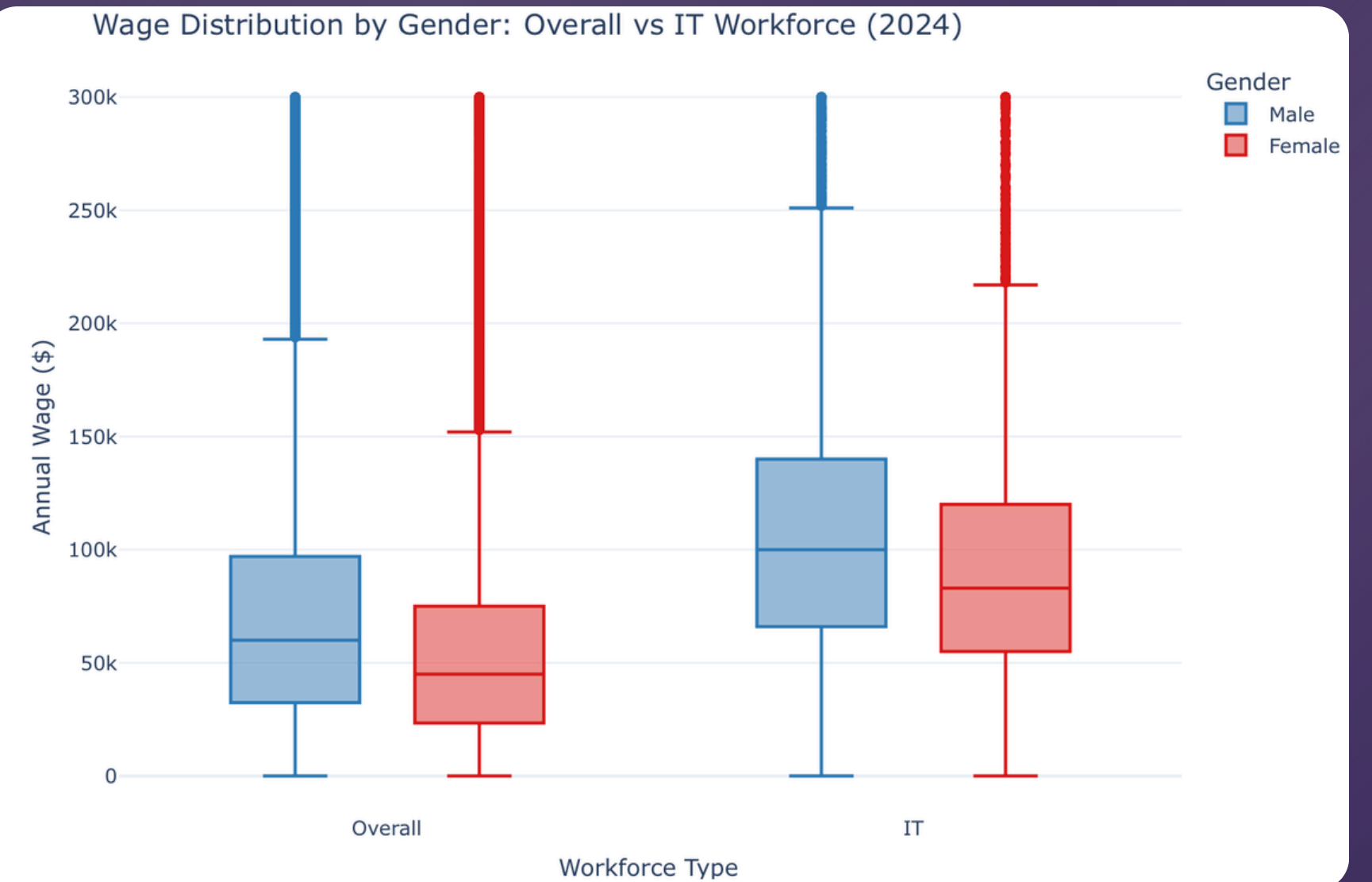


Visuals

Average Wage by State and Gender

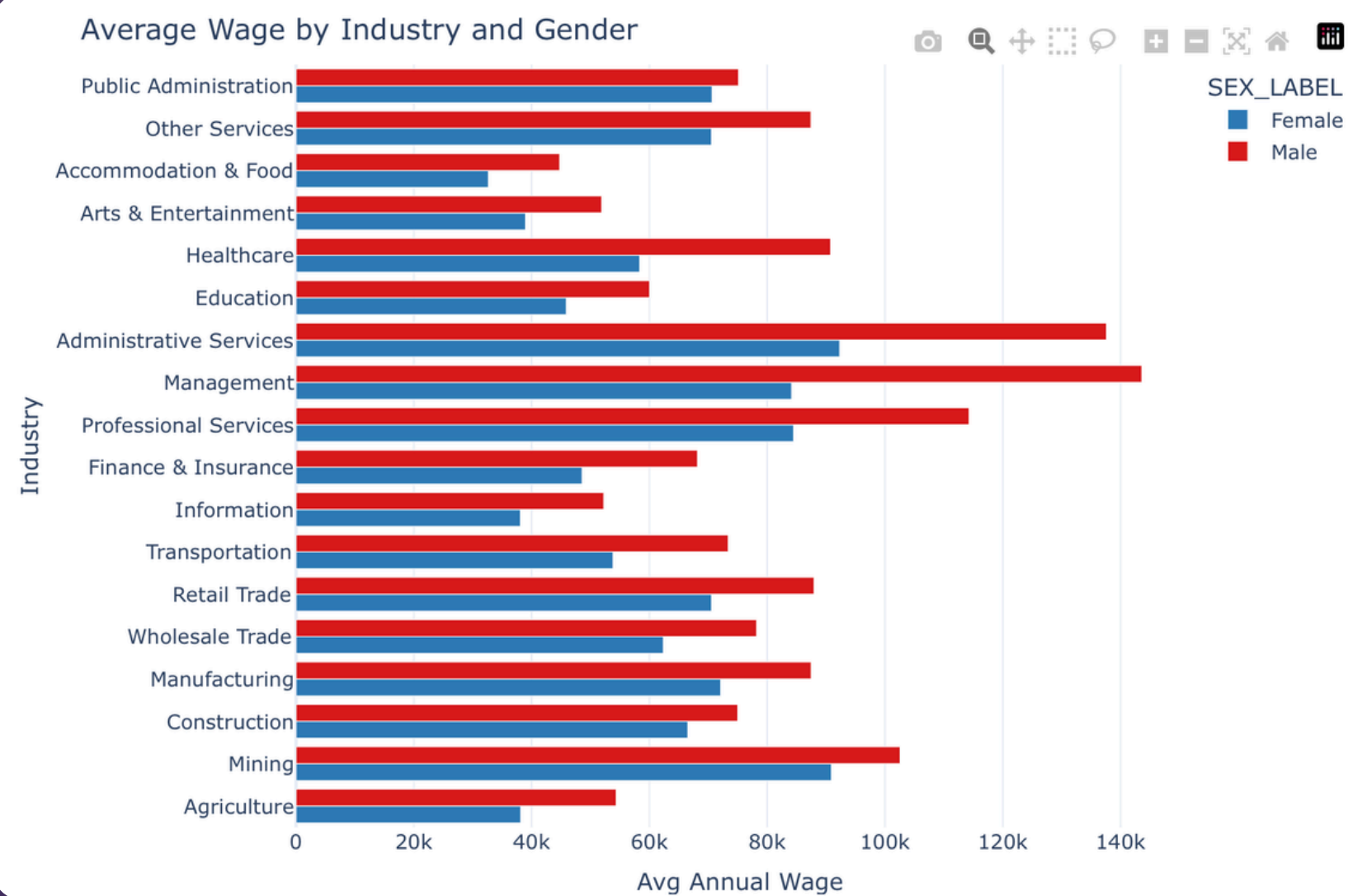


Wage Distribution by Gender

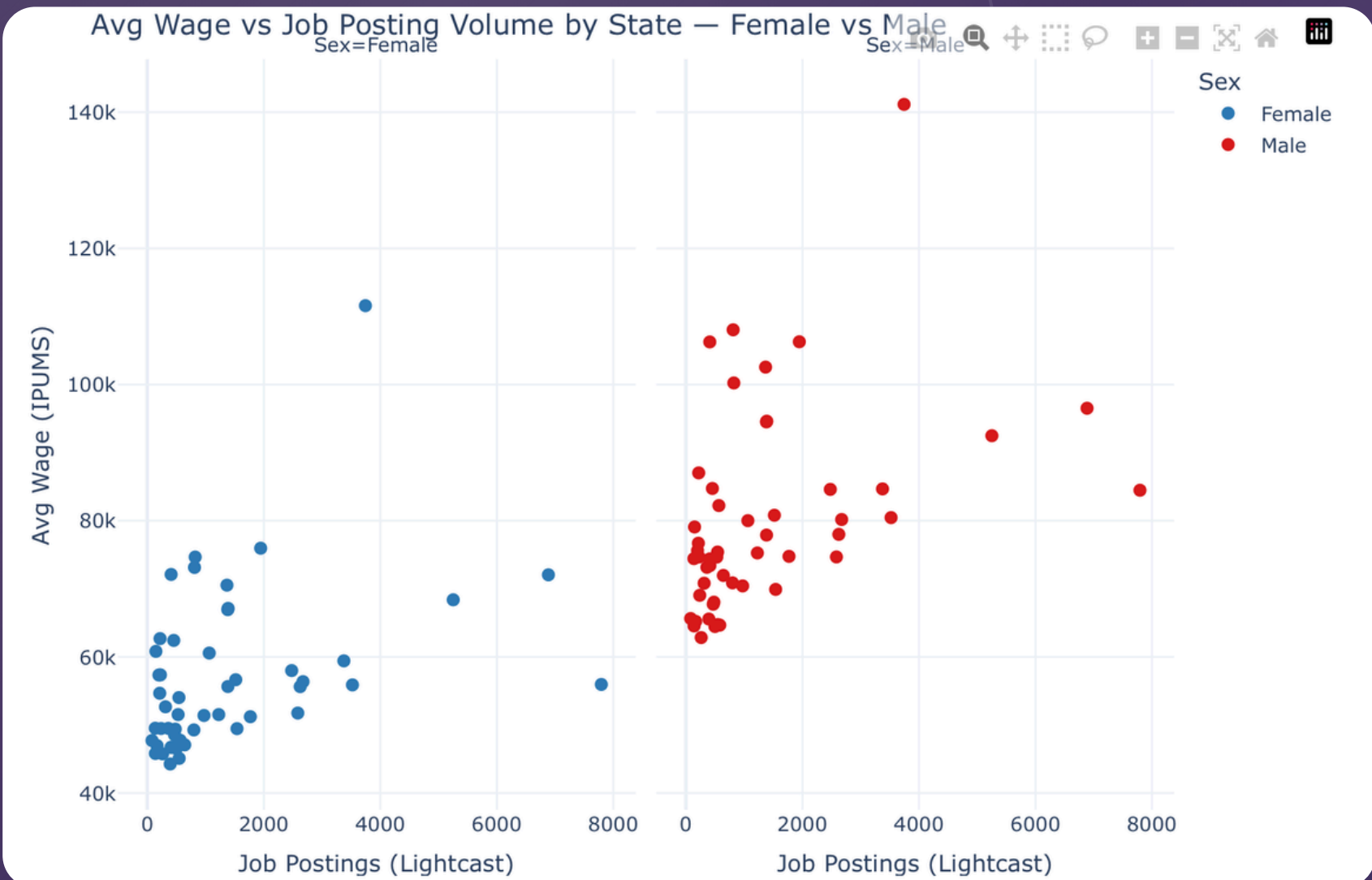


Visuals

Average Wage by Industry and Gender

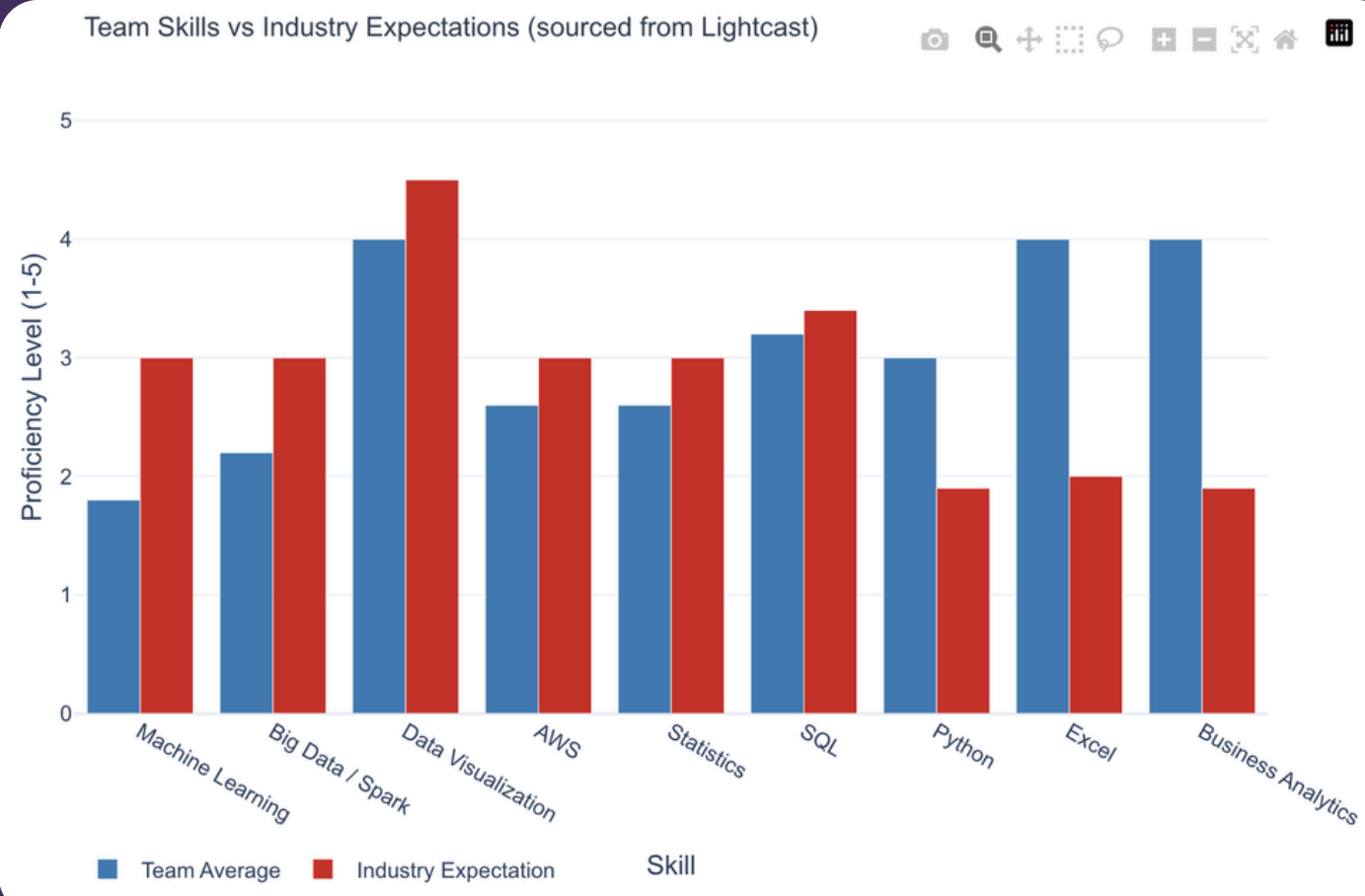
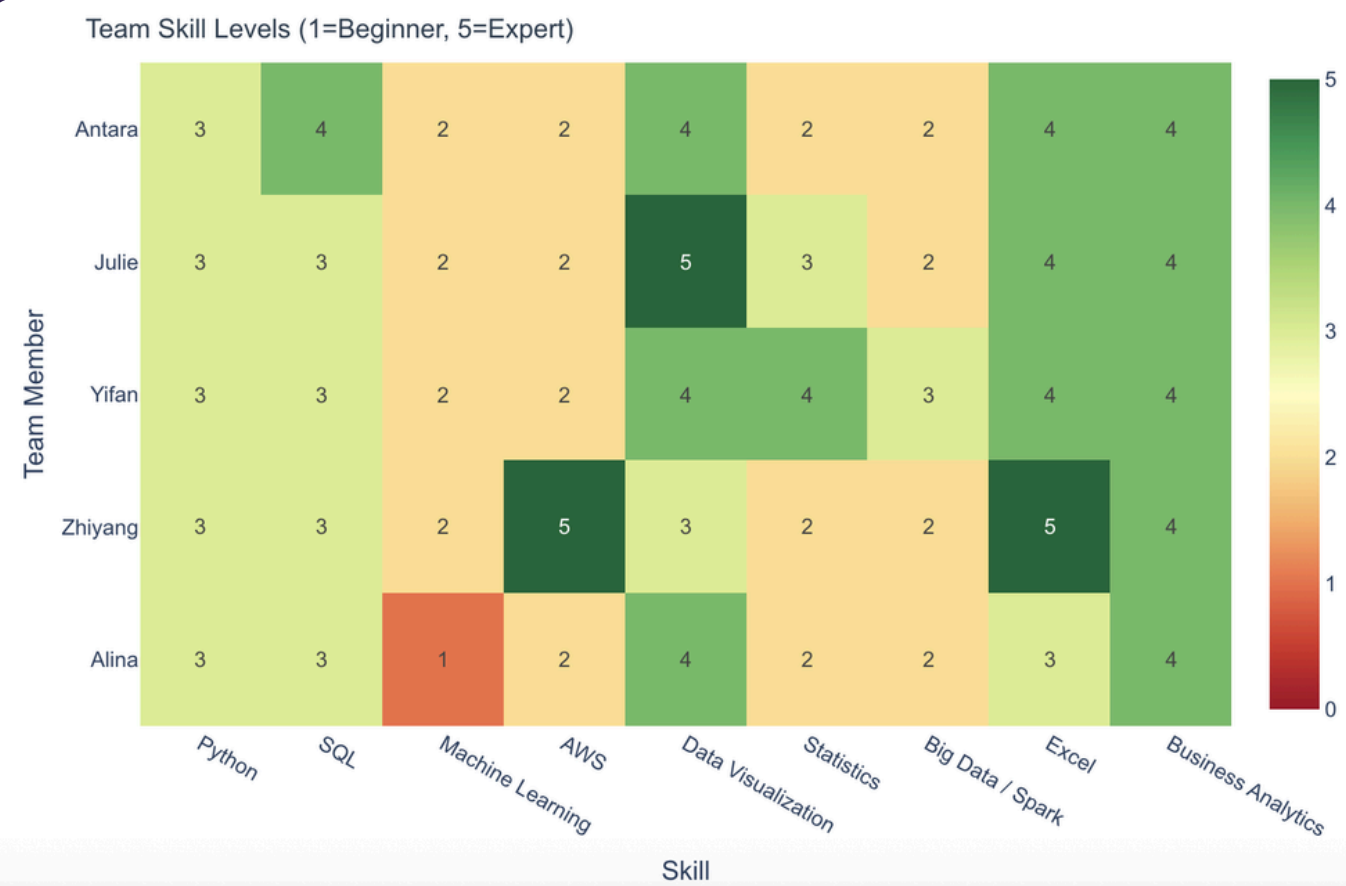


Average Wage vs Job Posting Volume by State



Skill Gap Analysis

Comparing our team's current capabilities with 2024 Lightcast IT market demand



Key Findings

- Team average meets basic requirements in SQL, Excel, and Business Analytics
- Strong specialization in AWS and Visualization
- Largest gaps exist in:
 - a. Machine Learning
 - b. Statistics
 - c. Spark / Big Data
- Soft skills like Communication and Problem Solving remain highly demanded in the market

Machine Learning

Clustering + Classification on IPUMS

KMeans (unsupervised) groups workers by wage and age into four pay tiers

Logistic regression predicts a worker's sex from age, wage, occupation and industry.

the female share falls as pay rises, from about **49%** → **23%** in the elite tier



Logistic ROC AUC:

0.83

gender is recoverable from a worker's economic profile alone.

Linear Regression Decomposition

To quantify exactly how much of the wage gap is explained by occupation/industry vs. unexplained

Baseline model: Age + Race + State + Sex

Conclusion: The gender wage gap is not primarily a story of women choosing lower-paying jobs

Raw Gap

\$24,749/year

Controlling for age, race and state only

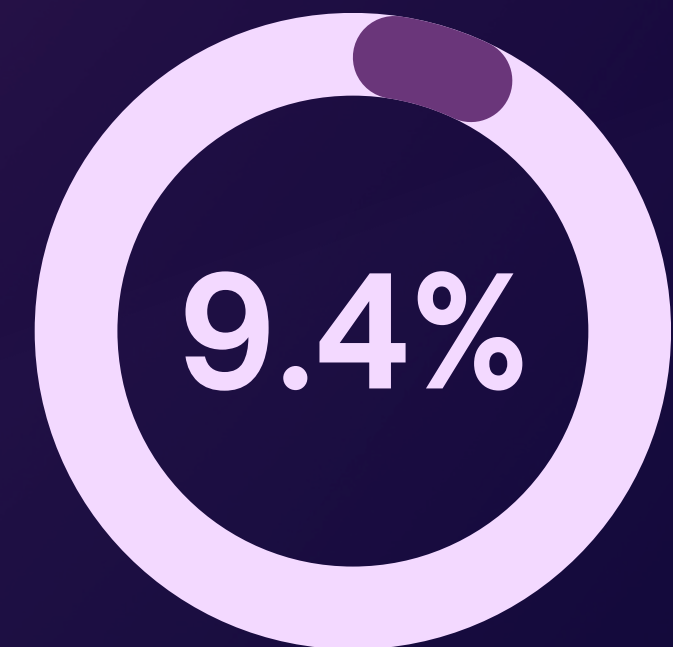
Adjusted Gap

\$22,426/year

Adding occupation and industry



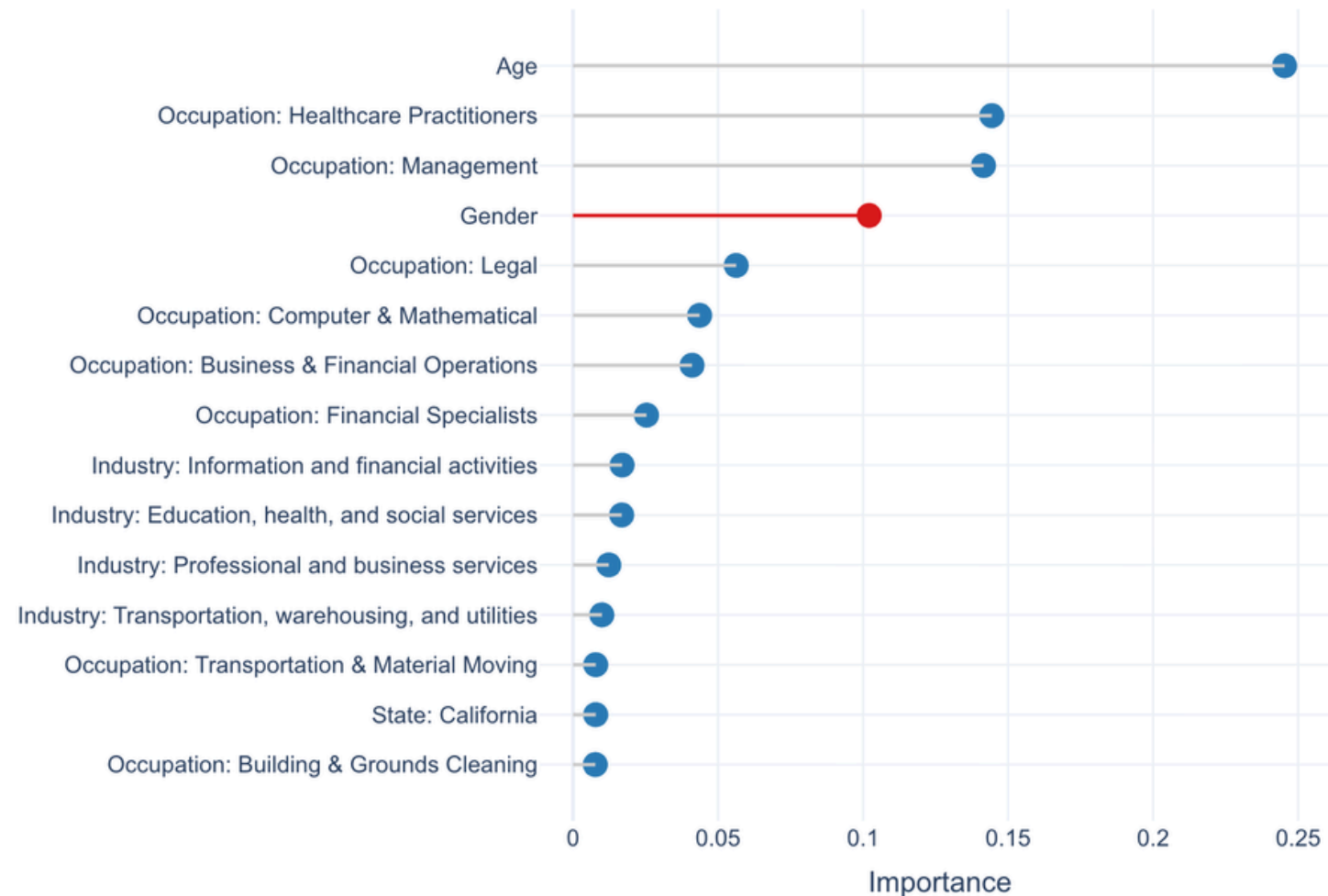
Unexplained



Explained

Random Forest

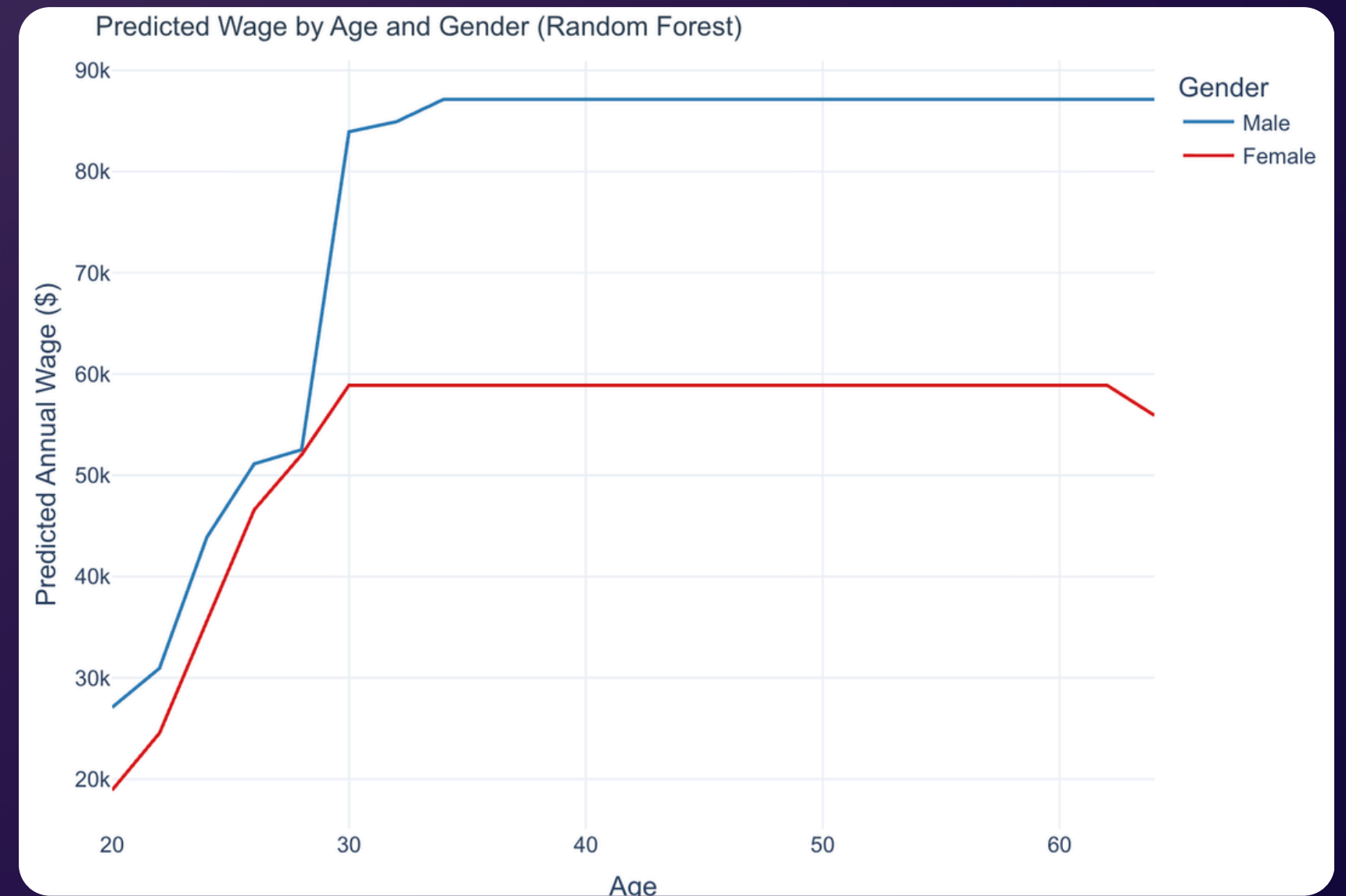
PySpark Random Forest Feature Importance (Top 15 of 172 features)



100 trees | Max depth 10 | 200K row sample

Parallelism : 2 cores | Partitions : 8

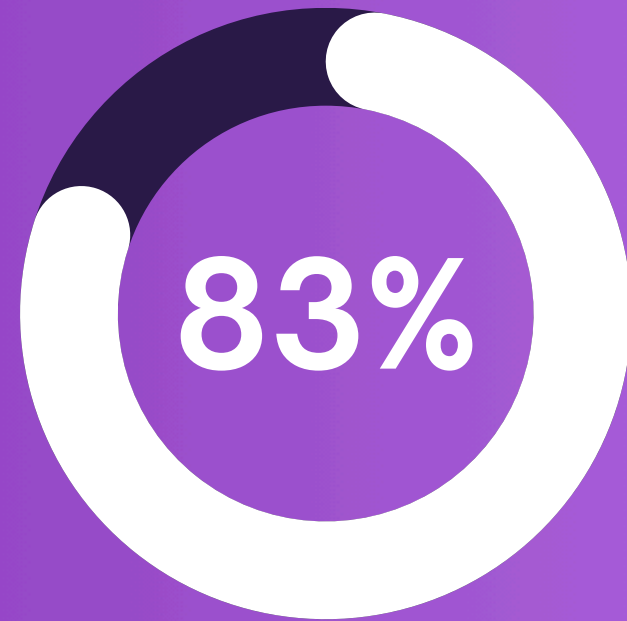
Train R^2 : 0.270 | Test R^2 : 0.236



Key Finding

Even when the model has access to occupation, industry, state and race - gender still ranks #4 of 172 features. This is not a regression artifact; it is a pattern the data independently discovered

Conclusion



0.83 ROC AUC: sex
predictable from economics
alone. National pattern: 25%
(blue) vs 22% (red).

91%

of the gender wage gap
is NOT explained by
occupation or industry .

raw gap
~\$24,700/yr

23%

women in elite-pay tier



VS

77%

men in elite-pay tier





JOURNAL OF ECONOMIC LITERATURE

SEPTEMBER 2023, VOLUME LXI, NUMBER 3

Symposium on Race, Ethnicity, and Economic Justice

Introduction

Gerald Jaynes

Perspectives on Race/Ethnicity and Economic Justice

The Public Sector and Resource Allocation

Daniel Aaronson, Daniel Hartley, Bhashkar Mazumder, and Martha Stinson

The Long-Run Effects of the 1930s Redlining Maps on Children

Touss Monarrez and David Schönholzer

Dividing Lines: Racial Segregation across Local Government Boundaries

Francisca M. Antman and Kalena E. Cortes

The Long-Run Impacts of Mexican American School Desegregation

Discrimination and Affirmative Action

Even K. Rose

A Constructivist Perspective on Empirical Discrimination Research

Shari J. Eli, Trevon D. Logan, and Boriana Miloucheva

The Enduring Effects of Racial Discrimination on Income and Health

Rajiv Sethi and Rohini Sonunathan

Meritocracy and Representation

Inequality and Income Support

Charlotte Cavaillé and Karine Van der Straeten

Immigration and Support for Redistribution: Lessons from Europe

Andria Snuythe and Linchi Hsu

The Minimum Wage as a Tool for Racial Economic Justice

Nathan Deutscher and Bhashkar Mazumder

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Antony Millner and Geoffrey Heal

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P. J. Gandon, Ken Kuttner, Sandeep Mazumder, and Caleb Stroup

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Kevin D. Hoover and Andrej Štorenčik

Who Runs the AEA?

Dionissi Aliprantis

Making Our Neighborhoods, Making Our Selves: A Review Essay

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Mordecai Kurz's *The Market Power of Technology: Understanding the Second Gilded Age* by Abhinav Gupta

Aime L. Murphy's *Virtuous Bankers: A Day in the Life of the Eighteenth-Century Bank of England* by Matthew Jaremski

Blau & Kahn (2017)



■ Sources: Blau & Kahn (2017); Goldin (2014); Arulampalam, Booth & Bryan (2007); Kleven, Landais & Søgaard (2019).

Thank You

Questions?